CROZ

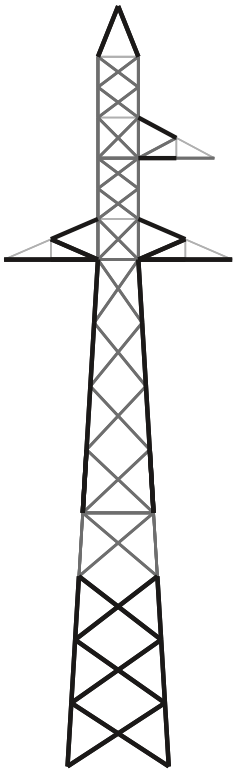
LATTICE TOWER STATIC CHECK

project	Tekla Open API - structural verification PoC
subject	Single-circuit 110 kV-type suspension tower, 30 m
model	test.db1 (C:\TeklaStructuresModels\test), 240 parts exported 2026-10-10T18:39:03
date	2026-10-10
status	Proof of concept. Simplified actions and checks; not reviewed by a licensed engineer. Not for construction.

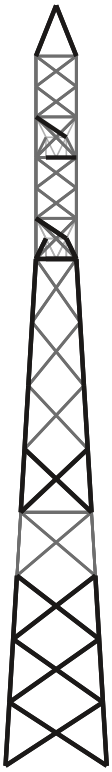
RESULT

PASS

Maximum utilisation	0.99	TWR_ARMBRACE_L1 L60*6 (slenderness 199/200, W225)
Members checked	240	372 FE elements, 136 nodes, second-order (P-Δ) analysis
Steel weight	5.17 t	Tekla reports 5.22 t (difference = root radii, which the analysis ignores)
Apex deflection	85 mm	largest over all combinations (factored loads)



Transverse elevation (X-Z)



Longitudinal elevation (Y-Z)

Line weight shows utilisation: light < 0.5, grey 0.5-0.8, black ≥ 0.8, dashed > 1.0.

MEMBER GROUPS

Group	No.	Profiles	Max util.	Governing check	Comb.	Strength util.
Legs	56	L80*8, L100*10, L120*12	0.85	slenderness 128/150	W000	0.78
Body diagonals	112	L50*5, L70*7, L80*8	0.93	slenderness 187/200	U270	0.33
Horizontals	24	L50*5, L80*8	0.71	slenderness 178/250	W135	0.50
Plan bracing	8	L50*5, L70*7	0.59	slenderness 147/250	W000	0.20
Peak	4	L90*9	0.87	slenderness 130/150	W000	0.06
Crossarm bottom chords	6	L80*8	0.80	slenderness 120/150	W000	0.27
Crossarm top chords	6	L60*6	0.50	slenderness 174/350 (tension only)	-	0.19
Crossarm bracing	24	L60*6	0.99	slenderness 199/200	W225	0.03

MOST UTILISED MEMBERS

ID	Name	Profile	Grade	z (m)	N_{Ed} (kN)	N_{Rd} (kN)	Util.	Check	Comb.	Bolts
60229	TWR_ARMBRACE_L1	L60*6	S235JR	21.20	0.1	-	0.99	slenderness 199/200	W225	1 x M
60529	TWR_ARMBRACE_L3	L60*6	S235JR	21.20	0.1	-	0.99	slenderness 199/200	W225	1 x M
60204	TWR_ARMBRACE_L1	L60*6	S235JR	20.00	0.0	-	0.93	slenderness 187/200	W000	1 x M
60504	TWR_ARMBRACE_L3	L60*6	S235JR	20.00	0.7	-	0.93	slenderness 187/200	W045	1 x M
54954	TWR_DIAG_YN	L80*8	S235JR	1.25	1.6	-	0.93	slenderness 187/200	U270	1 x M
54979	TWR_DIAG_YN	L80*8	S235JR	1.25	0.3	-	0.93	slenderness 187/200	W270	1 x M
55004	TWR_DIAG_XP	L80*8	S235JR	1.25	1.1	-	0.93	slenderness 187/200	W000	1 x M
55029	TWR_DIAG_XP	L80*8	S235JR	1.25	1.1	-	0.93	slenderness 187/200	W000	1 x M
55054	TWR_DIAG_YP	L80*8	S235JR	1.25	0.3	-	0.93	slenderness 187/200	W090	1 x M
55079	TWR_DIAG_YP	L80*8	S235JR	1.25	1.6	-	0.93	slenderness 187/200	U090	1 x M

55104	TWR_DIAG_XN	L80*8	S235JR	1.25	1.2	-	0.93	slenderness 187/200	W180	1 x M
55129	TWR_DIAG_XN	L80*8	S235JR	1.25	1.2	-	0.93	slenderness 187/200	W180	1 x M
55404	TWR_DIAG_XN	L80*8	S235JR	3.75	0.3	-	0.88	slenderness 176/200	W000	1 x M
55429	TWR_DIAG_XN	L80*8	S235JR	3.75	0.3	-	0.88	slenderness 176/200	W000	1 x M
55254	TWR_DIAG_YN	L80*8	S235JR	3.75	0.3	-	0.88	slenderness 176/200	W090	1 x M
55279	TWR_DIAG_YN	L80*8	S235JR	3.75	0.8	-	0.88	slenderness 176/200	W090	1 x M
55304	TWR_DIAG_XP	L80*8	S235JR	3.75	0.5	-	0.88	slenderness 176/200	W000	1 x M
55329	TWR_DIAG_XP	L80*8	S235JR	3.75	0.5	-	0.88	slenderness 176/200	W000	1 x M
55354	TWR_DIAG_YP	L80*8	S235JR	3.75	0.8	-	0.88	slenderness 176/200	W270	1 x M
55379	TWR_DIAG_YP	L80*8	S235JR	3.75	0.3	-	0.88	slenderness 176/200	W270	1 x M
60829	TWR_ARMBRACE_L2	L60*6	S235JR	25.20	0.2	-	0.88	slenderness 176/200	W225	1 x M
60129	TWR_ARMBRACE_L1	L60*6	S235JR	20.40	1.2	-	0.87	slenderness 174/200	W000	1 x M
60329	TWR_ARMBRACE_L3	L60*6	S235JR	20.40	0.5	-	0.87	slenderness 174/200	W000	1 x M
60029	TWR_ARMBRACE_L1	L60*6	S235JR	20.40	1.1	-	0.87	slenderness 174/200	W000	1 x M
60429	TWR_ARMBRACE_L3	L60*6	S235JR	20.40	0.9	-	0.87	slenderness 174/200	W000	1 x M
59879	TWR_PEAK	L90*9	S235JR	29.00	4.0	-	0.87	slenderness 130/150	W000	1 x M
59904	TWR_PEAK	L90*9	S235JR	29.00	4.0	-	0.87	slenderness 130/150	W000	1 x M
59929	TWR_PEAK	L90*9	S235JR	29.00	0.7	-	0.87	slenderness 130/150	W090	1 x M
59854	TWR_PEAK	L90*9	S235JR	29.00	0.8	-	0.87	slenderness 130/150	W090	1 x M
56979	TWR_LEG	L100*10	S355JR	18.75	76.0	-	0.85	slenderness	W000	1 x M

N_{Ed} positive = compression. Bolts per end: grade 8.8, single shear, bearing with $e_1 = 1.5\ d_0$ (information only).

VERIFICATION OF THE ANALYSIS

1. Global equilibrium. Sum of factored applied loads against sum of support reactions, per combination.

Comb.	Description	ΣF_x	ΣF_y	ΣF_z	ΣR_x	ΣR_y	ΣR_z	Error %
W000	1.35 G + 1.5 wind 0 deg	87.1	0.0	-94.9	87.1	0.0	-94.9	0.0003
W045	1.35 G + 1.5 wind 45 deg	61.9	50.1	-94.9	61.9	50.1	-94.9	0.0002
W090	1.35 G + 1.5 wind 90 deg	0.0	63.5	-94.9	0.0	63.5	-94.9	0.0001
W135	1.35 G + 1.5 wind 135 deg	-61.9	50.1	-94.9	-61.9	50.1	-94.9	0.0002
W180	1.35 G + 1.5 wind 180 deg	-87.1	0.0	-94.9	-87.1	0.0	-94.9	0.0003
W225	1.35 G + 1.5 wind 225 deg	-61.9	-50.1	-94.9	-61.9	-50.1	-94.9	0.0002
W270	1.35 G + 1.5 wind 270 deg	-0.0	-63.5	-94.9	0.0	-63.5	-94.9	0.0001
W315	1.35 G + 1.5 wind 315 deg	61.9	-50.1	-94.9	61.9	-50.1	-94.9	0.0002
U000	1.0 G + 1.5 wind 0 deg (uplift)	87.1	0.0	-70.3	87.1	0.0	-70.3	0.0003
U045	1.0 G + 1.5 wind 45 deg (uplift)	61.9	50.1	-70.3	61.9	50.1	-70.3	0.0002
U090	1.0 G + 1.5 wind 90 deg (uplift)	0.0	63.5	-70.3	0.0	63.5	-70.3	0.0001
U135	1.0 G + 1.5 wind 135 deg (uplift)	-61.9	50.1	-70.3	-61.9	50.1	-70.3	0.0002
U180	1.0 G + 1.5 wind 180 deg (uplift)	-87.1	0.0	-70.3	-87.1	0.0	-70.3	0.0003
U225	1.0 G + 1.5 wind 225 deg (uplift)	-61.9	-50.1	-70.3	-61.9	-50.1	-70.3	0.0002
U270	1.0 G + 1.5 wind 270 deg (uplift)	-0.0	-63.5	-70.3	0.0	-63.5	-70.3	0.0001
U315	1.0 G + 1.5 wind 315 deg (uplift)	61.9	-50.1	-70.3	61.9	-50.1	-70.3	0.0002
I000	1.35 G + 1.5 ice + 1.5 reduced wind on ice 0 deg	59.8	0.0	-123.8	59.8	0.0	-123.8	0.0004
I045	1.35 G + 1.5 ice + 1.5 reduced wind on ice 45 deg	37.3	20.0	-123.8	37.3	20.0	-123.8	0.0002
I090	1.35 G + 1.5 ice + 1.5 reduced wind on ice 90 deg	0.0	25.4	-123.8	-0.0	25.4	-123.8	0.0000
I135	1.35 G + 1.5 ice + 1.5 reduced wind on ice 135 deg	-37.3	20.0	-123.8	-37.3	20.0	-123.8	0.0001
I180	1.35 G + 1.5 ice + 1.5 reduced wind on ice 180 deg	-59.8	0.0	-123.8	-59.8	0.0	-123.8	0.0003
I225	1.35 G + 1.5 ice + 1.5 reduced wind on ice 225 deg	-37.3	-20.0	-123.8	-37.3	-20.0	-123.8	0.0001
I270	1.35 G + 1.5 ice + 1.5 reduced wind on ice 270 deg	-0.0	-25.4	-123.8	-0.0	-25.4	-123.8	0.0000
I315	1.35 G + 1.5 ice + 1.5 reduced wind on ice 315 deg	37.3	-20.0	-123.8	37.3	-20.0	-123.8	0.0002
ACC1	1.0 G + 1.0 ice + broken conductor L1	0.0	15.0	-89.5	-0.0	15.0	-89.5	0.0000

Loads and reactions in kN; reactions shown with reversed sign so the columns should match.

2. Independent hand calculation (W000). Section cut at z = 11.32 m, where the X-bracing diagonals cross on the face centreline, so the legs alone resist

the overturning moment: M = 674.8 kNm, V = -57.7 kN, leg spacing b = 2.642 m.
N_{leg} = (|V|/4 + M/(2b)) / cosθ = **142.6 kN** · FE linear = **141.7 kN** (-0.6 %) · FE second-order = 142.1 kN (P-Δ adds +0.2 %).

3. Code-check functions are unit-tested against EN 10056-1 section tables and EN 1993-1-1 buckling curve values (`test_eurocode.py`, run before every analysis).

FOUNDATION REACTIONS

Leg	Max compression (kN)	Comb.	Max uplift (kN)	Comb.	Max shear (kN)
+X / +Y	247.6	W045	200.5	U225	32.4
+X / -Y	247.6	W315	200.5	U135	32.4
-X / +Y	240.8	W135	205.6	U315	32.1
-X / -Y	240.8	W225	205.6	U045	32.1

APEX DEFLECTION

Comb.	δ (mm)	H / δ
W000	85.0	353
W045	66.6	451
W090	46.7	642
W135	58.7	511
W180	75.5	397
W225	58.7	511
W270	46.7	642
W315	66.6	451
U000	83.7	358
U045	65.5	458
U090	46.6	644
U135	59.6	503
U180	76.7	391
U225	59.6	503
U270	46.6	644
U315	65.5	458
I000	80.4	373
I045	51.9	578
I090	21.2	1418
I135	32.9	911

I180	60.3	497
I225	32.9	911
I270	21.2	1418
I315	51.9	578
ACC1	15.6	1924

ACTIONS

Wind: $v_{b,0}$ = 25.0 m/s, terrain z_0 = 0.05 m, peak pressure per EN 1991-1-4 (c_o = 1). Conductors Al/St 240/40 (d = 21.9 mm), earthwire Al/St 50/30; wind span 300 m, weight span 350 m, G_c = 0.75, C_c = 1.0. Tower: G_t = 1.05, $C_t = 3.96(1 - 1.5\chi + \chi^2)$ per panel. Ice 15.0 N/m on conductors with wind × 0.4 on the iced diameter. Broken conductor L1: 0.6 × H_{max} longitudinal. Steel self-weight × 1.1 for bolts and gussets.

Case	ΣF_x (kN)	ΣF_y (kN)	ΣF_z (kN)
G	0.0	0.0	-70.3
ICE	0.0	0.0	-19.2
BW	0.0	15.0	0.0
W0	58.0	0.0	0.0
W45	41.3	33.4	0.0
W90	0.0	42.3	0.0
W135	-41.3	33.4	0.0
W180	-58.0	0.0	0.0
W225	-41.3	-33.4	0.0
W270	-0.0	-42.3	0.0
W315	41.3	-33.4	0.0
WI0	39.9	0.0	0.0
WI45	24.8	13.4	0.0
WI90	0.0	16.9	0.0
WI135	-24.8	13.4	0.0
WI180	-39.9	0.0	0.0
WI225	-24.8	-13.4	0.0
WI270	-0.0	-16.9	0.0
WI315	24.8	-13.4	0.0

Combinations

W000	1.35 G + 1.5 wind 0 deg
W045	1.35 G + 1.5 wind 45 deg

W090	1.35 G + 1.5 wind 90 deg
W135	1.35 G + 1.5 wind 135 deg
W180	1.35 G + 1.5 wind 180 deg
W225	1.35 G + 1.5 wind 225 deg
W270	1.35 G + 1.5 wind 270 deg
W315	1.35 G + 1.5 wind 315 deg
U000	1.0 G + 1.5 wind 0 deg (uplift)
U045	1.0 G + 1.5 wind 45 deg (uplift)
U090	1.0 G + 1.5 wind 90 deg (uplift)
U135	1.0 G + 1.5 wind 135 deg (uplift)
U180	1.0 G + 1.5 wind 180 deg (uplift)
U225	1.0 G + 1.5 wind 225 deg (uplift)
U270	1.0 G + 1.5 wind 270 deg (uplift)
U315	1.0 G + 1.5 wind 315 deg (uplift)
I000	1.35 G + 1.5 ice + 1.5 reduced wind on ice 0 deg
I045	1.35 G + 1.5 ice + 1.5 reduced wind on ice 45 deg
I090	1.35 G + 1.5 ice + 1.5 reduced wind on ice 90 deg
I135	1.35 G + 1.5 ice + 1.5 reduced wind on ice 135 deg
I180	1.35 G + 1.5 ice + 1.5 reduced wind on ice 180 deg
I225	1.35 G + 1.5 ice + 1.5 reduced wind on ice 225 deg
I270	1.35 G + 1.5 ice + 1.5 reduced wind on ice 270 deg
I315	1.35 G + 1.5 ice + 1.5 reduced wind on ice 315 deg
ACC1	1.0 G + 1.0 ice + broken conductor L1

Tower wind panels (wind normal to face)

z (m)	A_t (m²)	χ	C_t	q_p (Pa)
0.0 - 2.5	1.34	0.139	3.21	556
2.5 - 5.0	1.30	0.146	3.18	689
5.0 - 7.5	1.26	0.155	3.14	805
7.5 - 10.0	1.44	0.196	2.95	886
10.0 - 12.5	1.01	0.153	3.15	948
12.5 - 15.0	0.98	0.167	3.08	999
15.0 - 17.5	0.95	0.186	2.99	1043
17.5 - 20.0	1.01	0.230	2.80	1080
20.0 - 21.6	0.56	0.220	2.85	1108
21.6 - 22.8	0.39	0.204	2.91	1126
22.8 - 24.0	0.47	0.246	2.74	1140
24.0 - 25.6	0.56	0.220	2.85	1156
25.6 - 26.8	0.39	0.204	2.91	1171

26.8 - 28.0	0.47	0.246	2.74	1183

METHOD AND SIMPLIFICATIONS

- Geometry, profiles and grades are read from the Tekla model through the Open API. Parts are merged at shared ends and split where X-bracing crosses (connected at the crossing).
- 3D frame with rigid joints, pinned foundations, second-order analysis (PyNite P-Δ). Design uses axial forces only, which is normal practice for bolted lattice towers. Secondary bending is not checked.
- Angle properties from nominal dimensions without root radii (A within 1-3 % of EN 10056-1, conservative). Geometric-axis inertia is used for the frame stiffness.
- Tension: min(A f_y/γ_{M0}, net section). Legs 0.9 A_{net} f_u/γ_{M2}; bracing connected by one leg $\beta = 0.55$ (EN 1993-1-8 3.10.3).
- Compression: buckling curve b. Legs and chords use $\lambda_{eff} = \lambda$ between nodes (conservative against the EN 1993-3-1 Annex G reduction). Bracing uses EN 1993-1-1 BB.1.2. For crossed diagonals, the out-of-plane length is the segment length when the partner diagonal is in tension, otherwise the full length.
- Slenderness limits (common practice, cf. ASCE 10): legs and chords 150, bracing 200, redundant members 250, tension-only 350.
- Partial factors are EN 1990 defaults (1.35 / 1.5), used as conservative placeholders. EN 50341-1 with the national annex uses reliability-based factors that must replace them.
- Not covered: wind on insulators and the peak, dynamic response, fatigue, connection design beyond bolt counts, foundations design, clearances, construction and maintenance load cases, line angle and dead-end cases.

Lastovska 23, HR - 10000 Zagreb | OIB 86132384544 | Tel. 01 6184 831 | Fax. 01 6184 833 | e-mail: croz-info@croz.net | www.croz.net
Trgovački sud u Zagrebu, MBS 080440187 | Temeljni kapital: 13.272,28 EUR, uplaćen u cijelosti
Predsjednik uprave: Vjekoslav Jadrešić | Članovi uprave: Krešimir Mudrovčić i Davorin Mudrovčić
ŽR: Erste & Steiermärkische bank d.d., Rijeka, HR57 2402006 1100516964 · Privredna banka Zagreb d.d., Zagreb, HR54 2340009 1110206937 · Hrvatska poštanska banka d.d., Zagreb, HR82 2390001 1198003879 · Zagrebačka banka d.d., Zagreb, HR59 2360000 1101615116 · Raiffeisenbank Austria d.d. Zagreb, HR29 2484008 1106516075